



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

MATHEMATICAL LITERACY P2/WISKUNDIGE GELETTERDHEID V2

MAY/JUNE/MEI/JUNIE 2026

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

Symbol/Kode	Explanation/Verduideliking
MA	Method with accuracy/ <i>Metode met akkuraatheid</i>
MCA	Method with consistent accuracy/ <i>Metode met volgehoue akkuraatheid</i>
CA	Consistent accuracy/ <i>Volgehoue akkuraatheid</i>
A	Accuracy/ <i>Akkuraatheid</i>
C	Conversion/ <i>Herleiding</i>
S	Simplification/ <i>Vereenvoudiging</i>
RT	Reading from a table/graph/document/diagram/ <i>Lees vanaf tabel/grafiek/dokument/diagram</i>
SF	Correct substitution in a formula/ <i>Korrekte vervanging in 'n formule</i>
O	Opinion/Explanation/ <i>Opinie/Verduideliking</i>
P	Penalty, e.g. for no units, incorrect rounding off, etc./ <i>Penalising, bv. vir geen eenhede, verkeerde afronding, ens.</i>
NPR	No penalty for correct rounding/ <i>Geen penalising vir korrekte afronding nie</i>
NPU	No penalty for omitting unit, but wrong unit is penalised/ <i>Geen penalisinge indien die eenheid uitgelos is nie, maar wel indien 'n verkeerde eenheid gebruik word.</i>
AO	Answer only/ <i>Slegs antwoord</i>

**These marking guidelines consist of 16 pages./
Hierdie nasienriglyne bestaan uit 16 bladsye.**

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled) version.
- Consistent accuracy (CA) applies in ALL aspects of the marking guidelines; however it stops at the second calculation error.
- If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra item presented.
- Rounding is an independent mark.
- General principle of marking, if the candidate makes one mistake one mark is deducted.
- A conclusion mark can only be given if $\frac{1}{3}$ of the total marks for the sub-question have been awarded.
- No penalty for rounding (NPR) if the first decimal is correct.

LET WEL:

- As 'n kandidaat 'n vraag TWEE KEER beantwoord, sien slegs die EERSTE poging na.
- As 'n kandidaat 'n antwoord van 'n vraag doodtrek (kanselleer) en nie oordoen nie, sien die doodgetrekte (gekanselleerde) poging na.
- Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyne toegepas; dit hou egter op by die tweede berekeningsfout.
- Wanneer 'n kandidaat aflesings vanaf 'n grafiek, tabel, uitlegplan en kaart aanbied en ekstra antwoorde gee, penaliseer vir elke ekstra item.
- Afronding tel as 'n afsonderlike punt.
- Die algemene beginsel van nasien is, as 'n leerder een fout maak, word een punt afgetrek.
- 'n Gevolgtrekkingspunt kan slegs gegee word indien $\frac{1}{3}$ van die totale punte vir die subvraag toegeken is.
- Geen penalisering vir afronding (NPR) nie as die eerste desimaal korrek is.

QUESTION/VRAAG 1 [30 MARKS/30 PUNTE] ANSWER ONLY FULL MARKS			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
1.1.1	✓✓ A Metric system/Metrieke stelsel	2A correct concept (2)	M L1 E
1.1.2	✓✓ A Strip chart/Strookkaart	2A correct concept (2)	MP L1 E
1.1.3	✓✓ A Layout plan/Uitlegplan	2A correct concept (2)	MP L1 E
1.1.4	✓✓ A Probability/Waarskynlikheid	2A correct concept (2)	P L1 E
1.1.5	✓✓ A Diameter / Middellyn	2A correct concept (2)	M L1 E
1.2.1	7 ✓✓ RT	2RT reading from table (2)	M L1 E
1.2.2	50% ✓✓ RT Saturday/Saterdag OR/OF SAT./SA. ✓RT	2RT percentage 1RT day (3)	P L1 E

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
1.2.3	✓✓RT Wednesday/ <i>Woensdag</i> OR/OF WED. /WO	2RT reading from table (2)	M L1 E
1.2.4	✓MA 1,3 mm + 1,4 mm + 18 mm = 20,7 mm ✓A	1MA addition of values 1A simplification (2)	M L1 E
1.2.5	✓A ✓A Seven minutes to three in the afternoon <i>Sewe minute voor drie in die middag</i> ✓A OR/OF Fifty-three minutes past two p.m. ✓A <i>Drie en vyftig minute na twee n.m.</i>	1A correct words 1A afternoon (2)	M L1 E
1.3.1	✓✓A Elevation map <i>Hoogtekaart</i>	2A correct concept (2)	MP L1 E
1.3.2	1 750 m ✓✓RT	2RT reading from map [Accept answers more than 1700 m but less than 1780 m] (2)	MP L1 E
1.3.3	D ✓✓RT	2RT reading from map (2)	MP L1 E
1.3.4	✓RT ✓RT 16 km – 5 km = 11 km ✓A	1RT 16 km 1RT 5 km 1A simplification [Accept 10,8 – 11,2] (3)	MP L1 M
		[30]	

QUESTION 2 [28 MARKS/28 PUNTE]			
Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
2.1.1	✓✓ A Bar scale/Staafskaal	2A bar scale (2)	MP L1 E
2.1.2	R37 ✓✓ RT	2RT correct number (2)	MP L1 E
2.1.3	Hazyview ✓✓ A	2A correct destination (2)	MP L2 M
2.1.4	14 ✓✓ RT	2RT correct number (2)	MP L1 M
2.1.5	✓A Scale/Skaal 33 mm = 6 km ✓A Map distance/Kaartafstand 83 mm Real distance/Werklike afstand $= \frac{83}{33} \times 6 \text{ km}$ ✓MA $= 15,09... \text{ km}$ ✓CA $\approx 15 \text{ km}$ ✓R ✓A OR/OF Scale/Skaal 11 mm = 2 km Map distance/Kaartafstand 83 mm ✓A Real distance/Werklike afstand $= \frac{83}{11} \times 2 \text{ km}$ ✓MA $= 15,09... \text{ km}$ ✓CA $\approx 15 \text{ km}$ ✓R ✓A OR/OF 3,3 cm = 6 km 3,3 cm = 600 000 cm 1 : 181 818,18181 ✓A ✓MA $8,1 \text{ cm} \times 181,818181 = 1\,472\,727,272 \text{ cm}$ $= 14,7 \text{ km}$ ✓CA $\approx 15 \text{ km}$ ✓R	1A measured scale 1A map distance 1MA using the ratio 1CA simplification 1R correct rounding OR/OF 1A measured scale 1A map distance 1MA using the ratio 1CA simplification 1R correct rounding OR/OF 1A measured scale 1A map distance 1MA using the ratio 1CA simplification 1R correct rounding (5)	MP L3 M

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
2.2.1	11 ✓✓ A	2A correct number (2)	MP L1 E
2.2.2	Number on lawn/Getal op grasperk ✓ MA ✓ RT = 16 000 – (5 500 + 3 500) = 7 000 ✓ CA	1MA subtracting 1RT correct values (at least 2) 1CA simplification (3)	MP L2 E
2.2.3	NW OR/OF North-west/Noordwes ✓✓ A	2A correct direction (2)	MP L2 M
2.2.4	Distance/Afstand ✓ RT = $\frac{100}{3} \times 91,44$ cm ✓ MA = 3 048 cm ✓ CA = 30,48 m ✓ C OR/OF 3 feet = 91,44 cm 1 foot = 30,48 cm ✓ RT 100 feet = $100 \times 30,48$ cm ✓ MA = 3 048 cm ✓ CA = 30,48 m ✓ C	1RT 100 ft 1MA using the ratio 1CA simplification 1C conversion OR/OF 1RT 100 ft 1MA using the ratio 1CA simplification 1C conversion (4)	MP L2 D
2.2.5	Last row in ZZ/Laaste ry in ZZ ✓ MA = 20 + 23 ✓ RT = 43 ✓ CA INVALID / NIE GELDIG NIE ✓ O	1MA addition 1RT 23 1CA simplification 1O verification (4)	MP L4 M
		[28]	

QUESTION 3 [31 MARKS]			
Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
3.1.1	$D = (90 \text{ m} - 40,3 \text{ m}) \div 2$ $= 24,85 \text{ m} \quad \checkmark \text{CA}$	1MA subtract and dividing by 2 1CA simplification AO (2)	M L2 E
3.1.2	Area of one penalty box/ <i>Opp. van een doelgebied</i> $= 40,3 \text{ m} \times 16,5 \text{ m} \quad \checkmark \text{SF}$ $= 664,95 \text{ m}^2 \quad \checkmark \text{A}$ Area of soccer field/ <i>Oppervlakte van sokkerveld</i> $= 120 \text{ m} \times 90 \text{ m}$ $= 10\,800 \text{ m}^2 \quad \checkmark \text{A}$ Percentage/ <i>Persentasie</i> = $\frac{664,95}{10\,800} \times 100\%$ $\checkmark \text{MCA}$ $= 6,156944 \% \quad \checkmark \text{CA}$ OR/OF Area of penalty box ÷ area of soccer field/ <i>Opp. van doelgebied ÷ opp. van sokkerveld</i> $= \frac{40,3 \text{ m}}{120 \text{ m}} \times \frac{16,5 \text{ m}}{90 \text{ m}} \quad \checkmark \text{SF} \quad \checkmark \text{A} \quad \checkmark \text{A}$ Percentage/ <i>Persentasie</i> = $0,335833... \times 0,18333 \times 100\%$ $= 6,156944\% \quad \checkmark \text{CA}$	1SF substitution 1A simplification 1A simplification 1MCA correct fraction 1CA simplification OR/OF 1MCA correct fraction 1SF substitution 1A simplification 1A simplification 1CA simplification NPR (5)	M L3 M
3.1.3	$\text{Perimeter/Omtrek} = 2 \times 120 \text{ m} + 2 \times 90 \text{ m}$ $= 420 \text{ m} \quad \checkmark \text{A}$ $\text{Distance ran/Afstand afgelê} = 10 \times 420 \text{ m} = 4\,200 \text{ m}$ $= 4,2 \text{ km} \quad \checkmark \text{C}$ INVALID/ <i>NIE GELDIG NIE.</i> $\checkmark \text{O}$ OR/OF	1SF substitution in formula 1A simplification 1CA multiplying correct values 1C conversion 1O conclusion OR/OF	M L4 M

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
	Length distance for 10 laps/<i>Lengte-afstand vir 10 rondtes</i> $= 120 \text{ m} \times 2 \times 10$ $= 2\,400 \text{ m}$ ✓ A Width distance for 10 laps/<i>Breedte-afstand vir 10 rondtes</i> $= 90 \text{ m} \times 2 \times 10$ $= 1\,800 \text{ m}$ ✓ A Total distance/<i>Totale afstand</i> $= 2\,400 \text{ m} + 1\,800 \text{ m}$ $= 4\,200 \text{ m}$ ✓ CA Distance in km/<i>Afstand in km</i> $= 4\,200 \div 1\,000 = 4,2$ km ✓ C INVALID/NIE GELDIG NIE ✓ O OR/OF ✓ SF Perimeter/<i>Omtrek</i> $= 2 \times 120 \text{ m} + 2 \times 90 \text{ m}$ $= 420 \text{ m}$ ✓ A Conversion to m/<i>Herleiding na m:</i> $4 \text{ km} = 4\,000 \text{ m}$ ✓ C Distance for 1 lap/<i>Afstand vir 1 rondte</i> $= 4\,000 \text{ m} \div 10$ $= 400 \text{ m}$ ✓ CA INVALID/NIE GELDIG NIE ✓ O	1A correct values for length 1A correct values for width 1CA simplification 1C conversion 1O conclusion OR/OF 1SF substitution in formula 1A simplification 1C conversion 1CA dividing correct values 1O conclusion (5)	
3.2.1	$P = \frac{1}{18}$ ✓ A $\approx 0,056$ ✓ R	1A numerator 1A denominator 1R correctly rounded (3)	P L2 E
3.2.2	Diameter/<i>Middel lyn</i> $= 1,2 \text{ m} \times 2 = 2,4 \text{ m}$ ✓ MA Total distance/<i>Totale afstand</i> $= 3,142 \times 2,4 \text{ m}$ ✓ SF $= 7,5408 \text{ m}$ ✓ CA OR/OF ✓ MA Total distance/<i>Totale afstand</i> $= 3,142 \times 1,2 \text{ m} \times 2$ ✓ SF $= 7,5408 \text{ m}$ ✓ CA	1MA calculate diameter 1SF correct substitution 1CA simplification OR/OF 1MA calculate diameter 1SF correct substitution 1CA simplification (3)	M L2 M
3.2.3	Direct/ <i>Direkte</i> ✓✓ A	2A direct	M L1 E

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
3.2.4 (a)	Radius = 5 m ÷ 2 = 2,5 m ✓ MA Volume = 3,142 × (2,5 m)² × 0,21 m = 4,123875 m³ ✓ SF ✓ C	1MA radius 1SF substitution of radius 1C conversion to metres (3)	M L2 M
3.2.4 (b)	Litres per minute = 5 × 3,78541178 ✓ MA = 18,9270589 ℓ ✓ A Volume = 4,123875 m³ × 1 000 = 4 123,875 ℓ ✓ C Number of minutes/Aantal minute = $\frac{4\ 123,875\ \ell}{18,9270589\ \ell/m}$ ✓ MCA = 217,8825...min ✓ CA Time/Tyd = 217,8825... ÷ 60 ✓ MCA = 3,63... ✓ CA = 3 hours/uur 38 minutes/minute ✓ R OR/OF Volume = 4,123875 m³ × 1 000 = 4 123,875 ℓ ✓ C Number of gallons/Aantal gellings = $\frac{4\ 123,875}{3,78541178}$ ✓ MCA = 1 089,413844 ✓ CA Number of minutes/Aantal minute = 1 089,413844 ÷ 5 ✓ MA = 217,882... ✓ CA ✓ MCA Time in hours and min/Tyd in ure en min = 217,882 ÷ 60 = 3,63... ✓ CA = 3 hours/uur 38 minutes/minute ✓ R OR/OF Number of litres/min/Aantal liters/min ✓ MA = 5 × 3,7841178 ℓ = 18,9270589 ℓ ✓ A Volume = 18,9270589 ℓ ÷ 1000 = 0,0189271 m³ ✓ C Number of minutes/Aantal minute = $\frac{4\ 123,875}{0,0189271}$ ✓ MCA = 217,8825047 ✓ CA Time/Tyd = 217,8825047 ÷ 60 ✓ MCA = 3 hours/uur 38 minutes/minute ✓ CA ✓ R	1MA multiplication 1A simplification litres 1C conversion 1MCA division 1CA simplification 1MCA divide minutes by 60 1CA simplification 1R rounding of minutes OR/OF 1C conversion 1MCA dividing 1CA simplification 1MA division by 5 1CA simplification 1MCA divide minutes by 60 1CA simplification 1R rounding of minutes OR/OF 1MA multiplication 1A simplification litres 1C conversion 1MCA division 1CA simplification 1MCA divide minutes by 60 1CA simplification 1R rounding of minutes (8)	M L3 D
			[31]

QUESTION 4 [33 MARKS]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
4.1.1	$\text{Radius} = \frac{20}{2} \text{ mm} \quad \checkmark \text{MA}$ $= 10 \text{ mm} \quad \checkmark \text{A} \quad \textbf{OR/OF} \quad 1 \text{ cm}$	1MA diameter by 2 1A simplification AO (2)	M L1 E
4.1.2	$\text{Height/Hoogte} = 309 \text{ mm} + 2(40 \text{ mm}) \quad \checkmark \text{C} \quad \checkmark \text{MA}$ $= 389 \text{ mm} \quad \checkmark \text{CA}$ $\approx 400 \text{ mm} \quad \checkmark \text{R}$	1C convert to mm 1MA addition 1CA simplification 1R rounding (4)	M L2 M
4.1.3	Total surface area ONE cork /Totale buite opp. EEN proppe $= 2 \times 3,142 \times (10 \text{ mm})^2 + 2 \times 3,142 \times 10 \text{ mm} \times 43 \text{ mm} \quad \checkmark \text{SF}$ $= 628,4 \text{ mm}^2 + 2\,702,12 \text{ mm}^2$ $= 3\,330,52 \text{ mm}^2 \quad \checkmark \text{CA}$ TSA for 3 layers on 53 corks /BO vir 3 lae op 53 proppe $= 3\,330,52 \text{ mm}^2 \times 53 \times 3 \quad \checkmark \text{MCA}$ $= 529\,552,68 \text{ mm}^2$ Convert to m²/Herlei na m² = $529\,552,68 \text{ mm}^2 \div 1\,000\,000$ $\checkmark \text{C}$ $= 0,52955... \text{ m}^2$ Paint needed/Verf benodig : $\frac{0,52955 \text{ m}^2}{3 \text{ m}^2} \times 1 = 0,176... \ell \quad \checkmark \text{MCA}$ $\approx 0,18 \ell$ $\therefore 0,18 \ell \times 1\,000 = 180 \text{ m}\ell \quad \checkmark \text{C}$ Total cost/Totale koste = $2 \times \text{R}55,00$ $= \text{R}110,00 \quad \checkmark \text{CA}$ OR/OF	CA from 4.1.1 1SF substitution in formula 1CA simplification 1MCA multiply by 53 1MCA multiply by 3 1C conversion to m ² 1MCA divide by 3 to get the number of litres 1C conversion to mℓ 1CA simplification OR/OF	M/F L3 D

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
	Total surface area ONE cork /Totale buite opp. EEN prop ✓SF $SA = 2 \times 3,142 \times (1 \text{ cm})^2 + 2 \times 3,142 \times 1 \text{ cm} \times 4,3 \text{ cm}$ $= 33,3052 \text{ cm}^2 \quad \checkmark \text{ CA}$ SA of 53 corks/BO van 53 proppe = $33,3052 \text{ cm}^2 \times 53$ ✓MC $= 1\,765,1756$ Convert to m/Herlei na m = $\frac{1\,765,1756}{10\,000}$ ✓C $= 0,17651756 \text{ m}^2$ Paint needed(1 coat)/Verf benodig(1 laag) $3 \text{ m}^2 = 1 \text{ litre}$ $0,17651756 \text{ m}^2 = x \text{ litres}$ $\frac{3x}{3} = \frac{0,17651756}{3}$ $x = 0,0588391 \text{ litres} \quad \checkmark \text{ MCA}$ Convert to ml/Herlei na ml = $0,0588391 \times 1\,000$ $= 58,8391 \quad \checkmark \text{ C}$ Total amount of paint/Totale hoeveel. verf = $58,8391 \times 3$ ✓MCA $= 176,51756 \text{ ml}$ Number of cans of paint/Aantal blikke verf $= 176,51756 \text{ ml} \div 125 \text{ ml}$ $= 1,41214$ ∴ Total cost/Totale koste = $2 \times \text{R}55$ $= \text{R}110 \quad \checkmark \text{ CA}$ OR/OF Total surface area ONE cork /Totale buite opp. EEN prop ✓SF ✓C $= [2 \times 3,142 \times (0,01 \text{ m})^2] + 2 \times 3,142 \times 0,01 \text{ m} \times 0,043 \text{ m}$ $= 0,00333052 \text{ m}^2 \quad \checkmark \text{ CA}$ TSA 53 corks/TBO 53 proppe = $0,00333052 \text{ m}^2 \times 53 \times 3$ ✓MCA $= 0,52955268 \text{ m}^2 \quad \checkmark \text{ CA}$ Number of litres/Aantal liters = $\frac{0,52955268 \text{ m}^2}{3}$ ✓MCA $= 0,17651756 \text{ litres} \times 1\,000$ $= 176,51756 \text{ ml} \quad \checkmark \text{ C}$ Number of cans/Aantal blikke = $\frac{176,51756 \text{ ml}}{125 \text{ ml}}$ $= 1,412$ Cost/Koste = $2 \times \text{R}55$ $= \text{R}110 \quad \checkmark \text{ CA}$ OR/OF	1SF substitution in formula 1CA simplification 1MCA multiply by 53 1C conversion to m² 1MCA divide by 3 to get the number of litres 1C conversion to ml 1MCA multiply 3 1CA simplification OR/OF 1SF substitution in formula 1C conversion to m 1CA simplification 1MCA multiply by 53 and 3 1CA simplification 1MCA divide by 3 to get the number of litres 1C conversion to ml 1CA simplification OR/OF	

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
	Total surface area ONE cork /Totale buite opp. EEN prop $\checkmark$ SF $= 2 \times 3,142 \times (10 \text{ mm})^2 + 2 \times 3,142 \times 10 \text{ mm} \times 43 \text{ mm}$ $= 628,4 \text{ mm}^2 + 2\,702,12 \text{ mm}^2$ $= 3\,330,52 \text{ mm}^2 \checkmark$ CA $= 3\,330,52 \div (1\,000)^2 = 0,00333052 \text{ m}^2 \checkmark$ C TSA for 3 layers/TBO vir 3 lae $= 0,00333052 \text{ m}^2 \times 3$ $= 0,009992 \text{ m}^2 \checkmark$ MCA TSA(53 corks)/TBO(53 proppe) $= 0,009992 \text{ m}^2 \times 53$ $= 0,52955 \dots \text{ m}^2 \checkmark$ MCA Paint/Verf: 1 ℓ covers/dek 3 m^2 1 tin/blik = 125 ml covers/dek $\frac{125 \text{ ml}}{1\,000 \text{ ml}} \times 3 \text{ m}^2 \checkmark$ C $= 0,375 \text{ m}^2 \checkmark$ MCA Number of cans/Aantal blikke $= \frac{0,52955 \text{ m}^2}{0,375 \text{ m}^2} = 1,41 \dots$ Total cost/Totale koste $= 2 \times \text{R}55,00$ $= \text{R}110,00 \checkmark$ CA OR/OF Radius $= 10 \text{ mm} = \frac{10}{1\,000} \text{ m} = 0,01 \text{ m}$ Height/Hoogte $= 43 \text{ mm} = \frac{43}{1\,000} \text{ m} = 0,043 \text{ m} \checkmark$ C Total Surface Area/Buite-oppervlakte $\checkmark$ SF $= 2 \times 3,142 \times (0,01)^2 + 2 \times 3,142 \times 0,01 \text{ m} \times 0,043 \text{ m}$ $= 0,00333052 \text{ m}^2 \checkmark$ CA Spread rate/Spreikoers $3 \text{ m}^2 / \ell$ Number of litre/Aantal liters $= \frac{0,00333052 \text{ m}^2}{3 \text{ m}^2} \times 1 \checkmark$ MCA $= 0,0011101733 \ell$ 3 layers/3 lae $= 0,0011101733 \text{ litres} \times 3 \checkmark$ MCA $= 0,0033305199 \text{ litres}$ For 53 corks/Vir 53 proppe $\checkmark$ C $= 0,0033305199 \times 1000 \text{ ml} \times 53 \checkmark$ MCA $= 176,5176 \text{ ml}$ Total cost/Totale koste $= 2 \times \text{R}55$ $= \text{R}110 \checkmark$ CA	1SF substitution in formula 1CA simplification 1C conversion to m^2 1MCA multiply by 3 1MCA multiply by 53 1C conversion ml to ℓ 1MCA coverage for 3 m^2 1CA simplification OR/OF 1C conversion to m 1SF substitution in formula 1CA simplification 1MCA divide by 3 to get the number of litres 1MCA multiply by 3 1C conversion to ml 1MCA multiply by 53 1CA simplification	(8)

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
4.2	Inside width of board/<i>Binnebreedte van bord</i> $= 37,2 \text{ cm} - 2(4 \text{ cm})$ ✓MA $= 29,2 \text{ cm}$ ✓CA $= 292 \text{ mm}$ ✓C Length of 4 patterns/<i>Lengte van 4 patrone</i> $= (43 \text{ mm} + 20 \text{ mm}) \times 4$ ✓MA $= 63 \text{ mm} \times 4$ $= 252 \text{ mm}$ ✓CA INVALID/NIE GELDIG NIE ✓O OR/OF Inside width of board/<i>Binnebreedte van bord</i> $= 37,2 \text{ cm} - 2(4 \text{ cm})$ ✓MA $= 29,2 \text{ cm}$ ✓CA Length of pattern/<i>Lengte van patroon</i> $= 43 \text{ mm} + 20 \text{ mm} = 63 \text{ mm}$ $= 6,3 \text{ cm}$ ✓C Number of repeats needed to complete the row/ <i>Getal patrone nodig om die ry te voltooi</i> $= \frac{29,2 \text{ cm}}{6,3 \text{ cm}}$ ✓MCA $= 4,63$ ✓CA INVALID/NIE GELDIG NIE. ✓O OR/OF ✓MA Board/<i>Bord</i>: $372 \text{ mm} - 80 \text{ mm} = 292 \text{ mm}$ ✓CA Space needed for exactly 4 repeats/<i>Ruimte benodig om presies 4 keer te herhaal</i> ✓MCA $= 292 \text{ mm} \div 4 = 73 \text{ mm}$ ✓CA Each pattern/<i>Elke patroon</i> $= 43 \text{ mm} + 20 \text{ mm}$ $= 63 \text{ mm}$ ✓A INVALID/NIE GELDIG NIE. ✓O	1MA subtraction 1CA simplification 1C conversion to mm 1MA multiplying by pattern 1CA simplification 1O opinion OR/OF 1MA subtraction 1CA simplification 1C conversion to cm 1MCA dividing by 6,3 1CA simplification 1O conclusion OR/OF 1MA subtraction 1CA simplification 1MCA dividing by 4 1CA simplification 1A simplification 1O conclusion (6)	MP L4 M
4.3.1	Number of lockers/<i>Aantal sluitkassies</i> $= 12 \times 5$ ✓MA $= 60$ ✓A OR/OF Number of lockers/<i>Aantal sluitkassies</i> $= 6 \times 10$ ✓MA $= 60$ ✓A	1MA multiplication 1A correct answer OR/OF 1MA multiplication 1A correct answer AO (2)	MP L1 E

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L															
4.3.2	To store extra books/lunch boxes/small sport bags/ extra clothes or shoes/ Keep learners' possessions safe. ✓✓O <i>Om ekstra boeke/kosblikke/klein sportsakke/ekstra klere of skoene te stoor/ Hou besittings van leerders veilig.</i>	2O reason (ACCEPT ANY ONE) (2)	MP L4 E															
4.3.3 (a)	Volume = L × w × h/Volume = L × b × h ✓SF 32 400 cm ² = 45 cm × B × 18 cm B = $\frac{32\,400\text{ cm}^3}{45\text{ cm} \times 18\text{ cm}}$ ✓MA = 40 cm ✓CA	1SF substitution 1MA changing subj of formula 1CA simplification (3)	M L2 M															
4.3.3 (b)	Total length of one stack/Totale lengte van een stapel ✓MCA ✓C (2 × 40 cm) + 5,7 cm + (2,8 cm × 2) = 91,3 cm ✓CA Total length of a set/Totale lengte van 'n stel ✓MCA ✓MA (91,3 cm × 5) + (4 × 0,2 cm) = 456,5 cm + 0,8 cm = 457,3 cm ✓CA OR/OF Length of one stack/ Lengte van een stapel ✓C ✓MCA = 2(400 mm) + 2(28 mm) + 57 mm = 913 mm ✓CA Total length of a set/Totale lengte van 'n stel = (913 mm × 5) + (4 × 2 mm) ✓MCA ✓MA = 4 565 mm + 8 mm = 4 573 mm ✓CA OR/OF Total lengths/Totale lengtes <table><tr><td></td><td>In cm</td><td>In mm</td></tr><tr><td>A:</td><td>2,8 × 10 ✓C</td><td>28 × 10 ✓MCA</td></tr><tr><td>B:</td><td>40 × 10 ✓MCA</td><td>400 × 10 ✓C</td></tr><tr><td>C:</td><td>5,7 × 5 ✓MCA</td><td>57 × 5 ✓MCA</td></tr><tr><td>Space/Spasie:</td><td>0,2 × 4 ✓MA</td><td>2 × 4 ✓MA</td></tr></table> ✓MCA Total/Totaal = 400 cm + 28 cm + 28,5 cm + 0,8 cm = 457,3 cm ✓CA OR/OF ✓MCA Total/Totaal = 4000 mm + 280 mm + 285 mm + 8 mm = 4 573 mm ✓CA		In cm	In mm	A:	2,8 × 10 ✓C	28 × 10 ✓MCA	B:	40 × 10 ✓MCA	400 × 10 ✓C	C:	5,7 × 5 ✓MCA	57 × 5 ✓MCA	Space/Spasie:	0,2 × 4 ✓MA	2 × 4 ✓MA	CA from 4.3.3 (a) 1MCA multiply by 2 1C conversion to cm 1CA length in cm 1MCA multiplication 1MA adding in between spaces 1CA simplification OR/OF 1C conversion to mm 1MCA multiply by 2 1CA length in cm 1MCA multiplication 1MA adding in between spaces 1CA simplification OR/OF 1C conversion to cm 1MCA multiply by 10 1MCA multiply by 5 1MA spaces in between 1MCA adding 1CA simplification OR/OF 1MCA adding 1CA simplification (6)	M L2 D
	In cm	In mm																
A:	2,8 × 10 ✓C	28 × 10 ✓MCA																
B:	40 × 10 ✓MCA	400 × 10 ✓C																
C:	5,7 × 5 ✓MCA	57 × 5 ✓MCA																
Space/Spasie:	0,2 × 4 ✓MA	2 × 4 ✓MA																
		[33]																

QUESTION 5 [28 MARKS]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
5.1.1	2 Feb ✓✓RT	2RT correct date (2)	M L1 M
5.1.2	Highest wave – Lowest wave/ <i>Hoogste golf – laagste golf</i> ✓RT ✓RT = 2,27 m – 0,09 m = 2,18 m ✓CA	1RT 2,27 1RT 0,09 1CA simplification (3)	M L2 M
5.1.3	Time duration/ <i>Tydsduur</i> : 06:46 to 17:25 ✓RT 06:46 to/tot 07:00 14 minutes/ <i>minute</i> 07:00 to/tot 12:00 5 hours/ <i>uur</i> ✓MA 12:00 to/tot 17:25 5 hours/ <i>uur</i> 25 minutes/ <i>minute</i> ✓CA Total time = 10 hours and 39 minutes <i>Totale tydsduur = 10 uur 39 min</i> INCORRECT/ <i>NIE KORREK NIE.</i> ✓O OR/OF Sunset time/ <i>Sonsondergangstyd</i> ✓RT = 6:46 + 10 hours/ <i>uur</i> 48 min = 16:46 + 48 min ✓MA = 17:34 ✓CA INCORRECT/ <i>NIE KORREK NIE.</i> ✓O OR/OF From/ <i>Vanaf</i> 06:46 to/tot 17:25 Duration/ <i>Tydsduur</i> = 17h25 – 06h46 ✓R = (17 × 60 min + 46 min) – (6 × 60 min + 25) = 1 045 min – 406 min = $\frac{639 \text{ min}}{60}$ ✓MA = 10 hours and 39 min ✓CA INCORRECT/ <i>NIE KORREK NIE</i> ✓O OR/OF 6:46 am to 5:25 pm = 17:25 ✓RT ∴ 17:25 – 6:46 ✓MA ✓CA = 10 hours 39 minutes / <i>10 ure 39 min</i> INCORRECT/ <i>NIE KORREK NIE</i> ✓O	1RT reading correct values 1MA finding duration 1CA simplification 1O conclusion OR/OF 1RT reading correct values 1MA adding duration 1CA simplification 1O comparison OR/OF 1RT reading correct values 1MA finding duration 1CA simplification 1O conclusion OR/OF 1RT reading correct values 1MA finding duration 1CA simplification 1O conclusion (4)	M L4 M

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
5.2.1	4 ✓✓A	2A correct number of airports (2)	MP L2 E
5.2.2 (a)	Length of the bar scale/Lengte van staafskaal: $= 20 \text{ mile/myl} \div 0,621371 \quad \checkmark C$ $= 32,1868\dots$ $\approx 32,2 \text{ km} \quad \checkmark CA$ OR/OF Scale in km/Skaal in km $3,5 \text{ cm} = 20 \text{ miles/myl} \div 0,621371 \quad \checkmark C$ $3,5 \text{ cm} = 32,18688\dots \quad \checkmark CA$ $1 \text{ cm} : 9,1962\dots \text{ km}$ $\approx 1 \text{ cm} : 9 \text{ km}$ OR/OF $1 \text{ km} = 0,621371 \text{ miles/myl}$ $1 \div 0,621371 = 1,6093444$ Length of the bar scale/Lengte van staafskaal: $1,609344498 \times 20 \quad \checkmark C$ $= 32,186888$ $\approx 32,2 \text{ km} \quad \checkmark CA$	1C convert to km 1CA simplification OR/OF 1C convert to km 1CA simplification OR/OF 1C convert to km 1CA simplification (2)	M L2 E
5.2.2 (b)	✓A 35 mm is 32,2 km 65 mm is n km $n = \frac{65}{35} \times 32,1868\dots \quad \checkmark MCA$ $= 59,7756\dots \text{ km}$ $\approx 59,8 \text{ km} \quad \checkmark CA$ VALID/GELDIG ✓O OR/OF If 5.2.2(a) scale is $1 \text{ cm} : 9,196254\dots \quad \checkmark CA$ Indien 5.2.2(a) skaal $1 \text{ cm} : 9,196254\dots$ is Distance/Afstand $= 6,5 \text{ cm} \times 9,196254\dots \quad \checkmark MCA$ $= 59,77\dots \quad \checkmark CA$ VALID/GELDIG ✓O	CA from 5.2.2(a) 1A measured distance 1MCA using the scale 1CA simplification 1O conclusion OR/OF 1CA measured distance from 5.2.2(a) 1MCA using the scale 1CA simplification 1O conclusion [Accept 33 mm – 36 mm] [Accept 58 km – 64 km] (4)	MP L4 M

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
5.3.1	Time /Tyd : 08:50 to/tot 9:21 ✓RT = 31 min ✓A Distance = Speed × Time/Afstand = <i>Spoed</i> × <i>Tyd</i> 21,3 miles/<i>myl</i> = Speed/<i>Spoed</i> × 31 min ✓SF ✓C 21,3 miles/<i>myl</i> = Speed/<i>Spoed</i> × 0,5166666... hours Speed in miles/hour/<i>Spoed in myl/uur</i> $= \frac{21,3}{0,516666667}$ = 41,22581 ✓CA	1RT correct times 1A duration 1SF substitution 1C conversion to hours 1CA simplification (5)	M L3 M
5.3.2	✓RT 25 miles/<i>myl</i> = 1 gallon/<i>gelling</i> ✓MA 21,3 miles × 0,04 gallon per mile = 0,852 gallons Total cost of fuel/<i>Brandstofkoste</i> ✓RT = 0,852 × \$4,757 ✓MA = \$4,05 ✓CA CORRECT/KORREK. ✓O OR/OF Fuel cost/<i>Brandstofkoste</i> \$4,757 per gallon /<i>gelling</i> SUV uses/<i>verbruik</i> 25 mpg ✓RT ✓RT Cost for/<i>Koste vir</i> 25 mile/<i>myl</i> = \$4,757 ✓MA ∴ 21,3 miles/<i>myl</i> = <i>n</i> $n = \frac{21,3}{25} \times \\$4,757$ ✓MA ≈ \$4,05 ✓CA CORRECT/KORREK. ✓O	1RT correct value 25 mpg 1MA number of gallons 1RT price per gallon 1MA multiply gallons with cost 1CA simplification 1O conclusion OR/OF 1RT price per gallon 1RT correct value 25 mpg 1MA number of gallons $\frac{21,3}{25}$ 1MA multiplication 1CA simplification 1O conclusion (6)	M L4 D
		[28]	
		TOTAL: 150	